Try MediChain in ten minutes

MediChain is live and ready to use. Everything you will see is fictional: the patients, records and staff are made up for judging, and no real patient data is stored.

Where

App	Address
Staff portal	https://13-247-143-188.sslip.io/doctor/
Patient app	https://13-247-143-188.sslip.io/patient/

Works on a phone or a computer, in any current browser. Passwords are in the entry form's demo-account fields, not in this document.

Good to know

- **One account per browser.** To be a doctor and a patient at once, use one normal window and one private (incognito) window, or two devices.
- The patient app is built for phones; the staff portal works on both.
- Emergency scanning needs camera permission when the browser asks.

Accounts

Role	Login	Who
Doctor	`naledi.mahlaba`	Dr Naledi Mahlaba, Emergency Medicine
Doctor	`riaan.botha`	Dr Riaan Botha, Internal Medicine
Nurse	`busisiwe.zulu`	Sister Busisiwe Zulu
Pharmacist	`fatima.ebrahim`	Fatima Ebrahim
Lab technician	`kagiso.mokoena`	Kagiso Mokoena
Paramedic	`bongani.mthembu`	Bongani Mthembu
Patient	`thabo.mahlangu`	Thabo Mahlangu: type 2 diabetes, hypertension, aspirin allergy
Patient	`zanele.khumalo`	Zanele Khumalo: asthma, folic acid
Patient	`johan.pretorius`	Johan Pretorius: atrial fibrillation on warfarin, penicillin allergy
Patient	`ayanda.ngcobo`	Ayanda Ngcobo: epilepsy, latex allergy

A suggested route

1. **As a patient** (Thabo, on a phone): open the dashboard, then **Emergency card**, the summary a responder needs. Open **Medical ID** and download the QR code; it opens as an image you can hold up to another device's camera.
2. **Access history**: every time staff opened Thabo's record, with who, when and why. Patients can see this; it is the heart of MediChain.
3. **As a paramedic** (Bongani, second device or private window): **Emergency access**, then **QR Code**, then **Scan QR code with camera**, and point it at Thabo's QR. Blood type, allergies and contacts appear in seconds, under a time-limited grant.

4. **Back as Thabo**: refresh **Access history**; the paramedic's emergency access is listed.
5. **As a doctor** (Dr Mahlaba): open **Patients**, then Thabo's chart. Write a SOAP note from a template, or message Thabo (**Messages**, then **Start a new conversation**). Thabo's open screen receives it live.
6. Optionally, the pharmacist (prescriptions to dispense) and lab technician (specimens and results) show the rest of the care loop.

If something does not work during judging, email **kkgawatlh9@gmail.com**. The site is monitored, with alerts on errors and failed sign-ins.

Your record goes with you. Every access is visible.

MediChain is a South African health record and emergency-access platform. A patient carries their emergency summary as a QR code or NFC card; a paramedic or clinician reads it in seconds on an approved device; and the patient sees every access to their record, by whom and why.

The problem

South African public facilities still depend heavily on paper folders. Studies in Limpopo found that missing records made patients wait longer and, in some cases, meant they were "treated without medical history" [1]. A 2024 review found that "numerous South African public health facilities fail to meet their record-keeping obligations as required by law" [2]. In an emergency, the information that matters most (allergies, medicines, chronic conditions, who to call) is often the hardest to find.

What MediChain does

- **Emergency access in seconds:** scan the patient's QR code with a phone or laptop camera, tap their NFC card on an Android phone, or type the card number. Access is time-limited, tied to an approved device and recorded.
- **Patients in control:** patients grant and withdraw staff access, sign or refuse consent, and see a full access history, including emergency access.
- **One record for the care team:** seven roles (patient, doctor, nurse, pharmacist, lab technician, paramedic, administrator) work from the same record: notes, orders, prescriptions, results, handovers and secure messages.
- **Works when the network does not:** if there is no connection, a scanned QR still shows the patient's own emergency summary, clearly labelled as unverified.

Where it stands

Live	7	1,045	276
on AWS in Cape Town, with HTTPS	roles, 116 screens	automated server tests passing	step role-journey check, every role

MediChain is a working proof of concept, built and run by one developer. It holds fictional data only: real patient data waits on legal and regulatory sign-off (POPIA authorisation and a registered Information Officer), which are tracked as release gates.

The ask

A clinic, hospital or EMS partner for a supervised pilot of one workflow, measuring how quickly staff find critical information and how well patients understand their access history.

Lukau Invasion (Pty) Ltd · 2026/600804/07 · founder-developer Keorapetswe Kgoatlha · 2nd place, Rust Africa Hackathon 2026

Sources

1. Marutha, N. S. & Ngoepe, M. (2017). The role of medical records in the provision of public healthcare services in the Limpopo province of South Africa. *South African Journal of Information Management*, 19(1), a873. https://doi.org/10.4102/sajim.v19i1.873
2. Kolawole, O. (2024). Non-compliant health record-keeping in South Africa: Judicial responses, and progress towards a digital remedy. *The African Journal of Information and Communication*, (33), 1–19. https://doi.org/10.23962/ajic.i33.18892

Who MediChain helps, and how we will know

The problem, in South African evidence

Finding	Source
Missing medical records "negatively affected timely and effective healthcare service delivery"; patients waited longer and some were "treated without medical history". Nurses and doctors were "unable to assist patients or treat them immediately".	Survey of 40 public hospitals, Limpopo [1]
"Numerous South African public health facilities fail to meet their record-keeping obligations as required by law", undermining patients' medical-negligence claims.	Review of South African case law, 2024 [2]
At Khayelitsha Hospital, stacks of un-scanned paper folders left months of emergency-centre records unavailable on the hospital's electronic system.	Single-hospital study, emergency centre, 2012–2013 [3]
National policy calls for person-centred digital health, a complete electronic health record and a common patient identifier.	National Digital Health Strategy 2019–2024 [4]

These studies describe the problem; they are not MediChain outcomes. The Khayelitsha study is historical and covers one hospital, so it is not a national rate.

Who benefits

Who	What changes
Patients	Carry their emergency summary (QR or NFC); see every access to their record; control who on the care team can open it; read their notes, results, medicines and messages.
Paramedics and emergency staff	See blood type, allergies, medicines, conditions and emergency contacts in seconds, on an approved device, with the access recorded.
Clinicians and nurses	One shared record for notes, orders, results and handovers; notifications only about their own patients.
Pharmacists and laboratories	Prescriptions and specimens tied to the patient and the ordering clinician; dispensing and results visible to the care team.
Facilities and regulators	A tamper-evident record of every access, which supports POPIA accountability and disputes about care.

What a pilot will measure

- **Time to critical information:** seconds from first contact to a responder seeing allergies and medicines, compared with the current folder or phone-call route.
- **Completeness at handover:** share of emergency handovers with allergies, medicines and contacts recorded.
- **Patient understanding:** whether patients can say who opened their record, and why, after using access history.
- **Adoption:** active staff per shift, and workflows completed without paper fallback.

Every number in a pilot report will be measured; MediChain makes no claim about mortality or error reduction.

Sources

1. Marutha, N. S. & Ngoepe, M. (2017). The role of medical records in the provision of public healthcare services in the Limpopo province of South Africa. *South African Journal of Information Management*, 19(1), a873. https://doi.org/10.4102/sajim.v19i1.873
2. Kolawole, O. (2024). Non-compliant health record-keeping in South Africa: Judicial responses, and progress towards a digital remedy. *The African Journal of Information and Communication*, (33), 1–19. https://doi.org/10.23962/ajic.i33.18892
3. Ohuabunwa, E. U., Sun, J., Jubanyik, K. J. & Wallis, L. A. (2016). Electronic medical records in low to middle income countries: the case of Khayelitsha Hospital, South Africa. *African Journal of Emergency Medicine*, 6(1), 38–43. https://doi.org/10.1016/j.afjem.2015.06.003
4. National Department of Health (2019). *National Digital Health Strategy for South Africa 2019–2024*. https://knowledgehub.health.gov.za/system/files/elibdownloads/2023-04/national-digital-strategy-for-south-africa-2019-2024-b.pdf

Built so patients can trust it

Health information is special personal information under POPIA. MediChain is designed so that every access has a reason, is limited to the people treating the patient, and leaves a record the patient can read.

Who can open a record

Route	How it works
The patient's own grant	A clinician requests access; the patient approves it for a set period (at most 30 days) and can withdraw it at any time.
A care relationship	An appointment, consultation or referral gives the treating clinician access for a limited window.
Emergency (break-glass)	Needs an active work context, an approved device and a reason; the server issues a time-limited grant. The patient sees the access in their history.
Everyone else	Refused. Holding a clinical role is not enough on its own. Administrators manage accounts and devices, not clinical records.

Patients see and decide

- **Access history:** every chart opening and emergency read, with who, when and the stated reason.
- **Consent with a legal basis:** consent is recorded as given or refused, never assumed; withdrawal is kept as a record, not deleted.
- **Notifications reach only the right people:** a patient's reminders go to that patient; clinical alerts go to the patient and the clinicians treating them; administrators receive security alerts only.

How data is protected

Control	Implementation
Passwords	Never sent when signing in: the browser sends a one-way proof (PBKDF2 then HKDF), stored as Argon2id. New passwords must be 15 or more characters and are checked against known-breach lists without sending the password.
Accounts	Authenticator-app (TOTP) step-up for privileged actions; lockout after repeated failures; sessions revocable.
Encryption	HTTPS with security headers and a strict content security policy; encrypted documents (ChaCha20-Poly1305); encrypted disks and backups.
Audit	Access-log and audit rows are batched into Merkle trees, hash-chained, signed (Ed25519) and exported to write-once (Object Lock) storage, so a past record of access cannot be quietly rewritten.
Emergency QR	Carries a server-signed proof, so a typed patient number cannot open emergency access.
Operations	Hosted in South Africa (AWS Cape Town); nightly backups with an automatic restore test; alerts on errors and failed sign-ins.

Honest status: real patient data is gated

MediChain holds fictional data only. Before real patients, seven release gates must close. The engineering gates are in progress; three are legal and organisational:

- a registered Information Officer for the operating company;
- POPIA prior-authorisation and cross-border processing assessments;
- sign-off by a South African health and privacy lawyer.

> Irreversible deletion of clinical records is deliberately not built: records are retired, not destroyed, until retention rules are approved.

How MediChain is built

Architecture

Layer	Technology
Patient app and staff portal	React 18, TypeScript, Vite; installable web apps (PWA) for phone and desktop
Server	Rust (Actix-web): 500+ endpoints; every request is checked against a verified session and the caller's role and relationship to the patient
Data	PostgreSQL 16; encrypted clinical documents stored in the database
Real time	Server-sent events, delivered only to the accounts an event concerns
Cryptography	ChaCha20-Poly1305, Argon2id, Ed25519-signed audit batches
Hosting	AWS af-south-1 (Cape Town): EC2 behind Caddy with automatic HTTPS; S3 for backups and write-once audit checkpoints; CloudWatch alarms

Emergency path

1. The responder picks the facility's approved device and opens a professional work context.
2. They scan the patient's QR (any camera: built-in detector where available, a bundled decoder elsewhere), tap an NFC card (Web NFC on Android), or type the card number.
3. The server checks the device, the work context and the card or signed QR proof, then issues a time-limited emergency grant.
4. The summary is shown; the disclosure is written to the audit stream, and the patient sees it in their access history.
5. With no connection, the QR's own printed summary is shown, labelled unverified and unrecorded.

Measured quality

1,045 server tests passing

855 staff-portal tests

280 patient-app tests

35 automated code checks

- **Role journeys:** 276 steps across all seven roles fill in each role's forms with the page's own data and read the result back as a different user. The one defect this found (consent signing) was fixed the same day.
- **Live check:** a script tests the live site: HTTPS certificate, security headers and content security policy, then signs in as every reviewer account and opens every screen it may reach. It passed 24 of 24 checks.
- **Browser tests:** real-browser tests cover emergency QR scanning with a camera, live messaging on a phone-sized screen, notification history and cross-role workflows.
- **Gates in CI** refuse, among others: an unauthorised endpoint, a form that saves without confirming, a form that sends values nobody entered, and unreadable colour contrast.

Clinical logic

Clinical scores and alerts (for example qSOFA, drug interactions and lab-trend flags) are computed on the server by explicit rules, so every result can be explained. MediChain does not use a predictive model, and an unmeasured value is shown as unknown, never as zero.

Test counts measured 20 October 2026.

A sustainable service, starting with one partner

Model

Element	Approach
Customer	Private clinics, hospital groups and EMS services first; public facilities through partnerships once POPIA authorisation is in place.
Revenue	A per-facility subscription, plus onboarding and support. Free for patients. Pricing is a hypothesis to validate with the first partner.
Running cost	One hosted instance in Cape Town serves the full product today, under a USD 60 per month budget alarm. Costs grow with facilities, not with patients.
Distribution	Web apps that install like apps, so no app-store dependency; an Android app is the next step for card emulation and offline use.

Position

Alternative	What it does well	Where MediChain differs
Paper folders	Familiar; no infrastructure	Available at the point of care, and every access recorded
Hospital record systems (e.g. OpenMRS, Bahmni)	Established clinical operations	Patient-visible access history and controlled emergency access, built in from the start
Emergency apps (e.g. iER)	Panic alerts and self-reported medical information	Clinician-maintained data behind the emergency summary, read on approved devices, with the read shown to the patient

Comparisons describe focus, not shortcomings: established platforms have their own audit and patient features.

Roadmap

Next	What
Pilot	One partner, one workflow (emergency summary and handover), synthetic training first, then an approved patient pilot.
Regulatory	Information Officer registration, POPIA prior authorisation, legal sign-off.
Product	Android app (phone-to-phone NFC), wider languages, a custom domain, write-once cloud audit sink in production mode.
Team	Today one founder-developer; first hires: a clinical lead and a second engineer, funded through grants (for example NYDA) and pilot revenue.

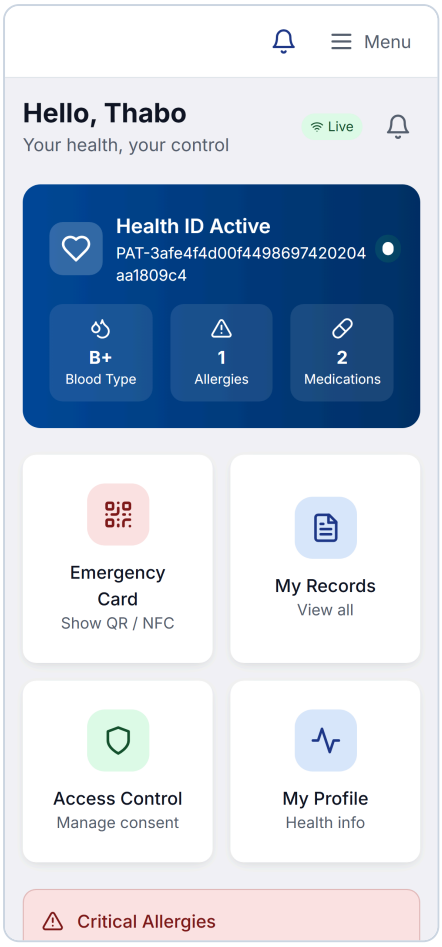
The ask

A clinic, hospital or EMS partner for a supervised pilot, and introductions to funders who back South African health technology.

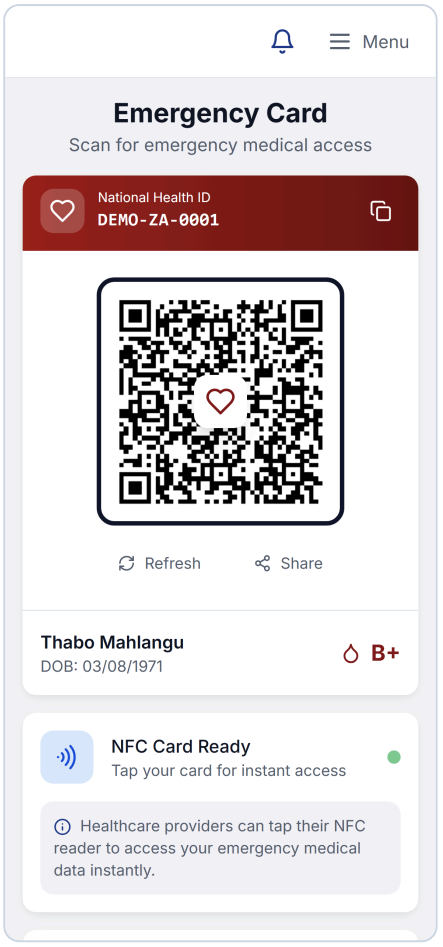
The product, as judges will see it

Captured from the live site with the fictional reviewer accounts.

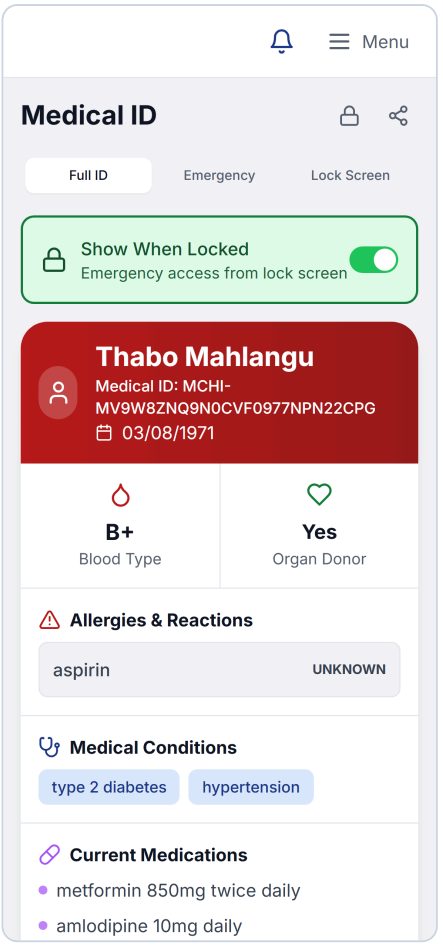
Patient app (phone)



Dashboard

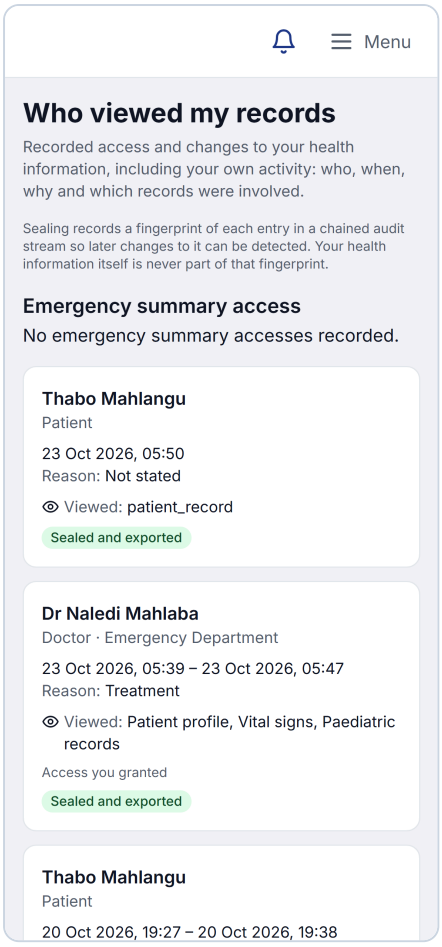


Emergency card

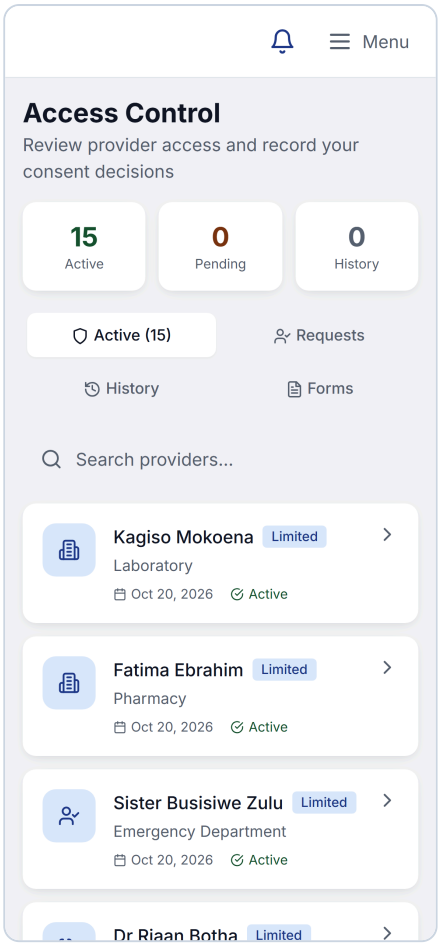


Medical ID and QR code

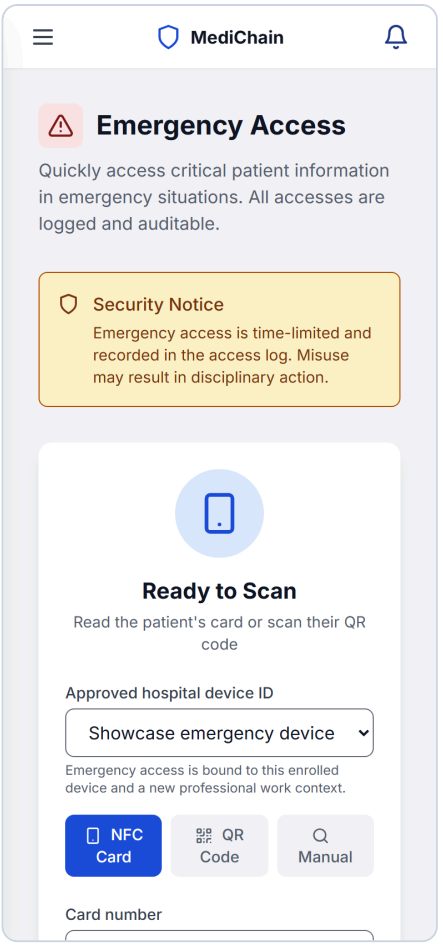
The patient's view of access, and the responder's phone



Access history: who opened the record, when and why

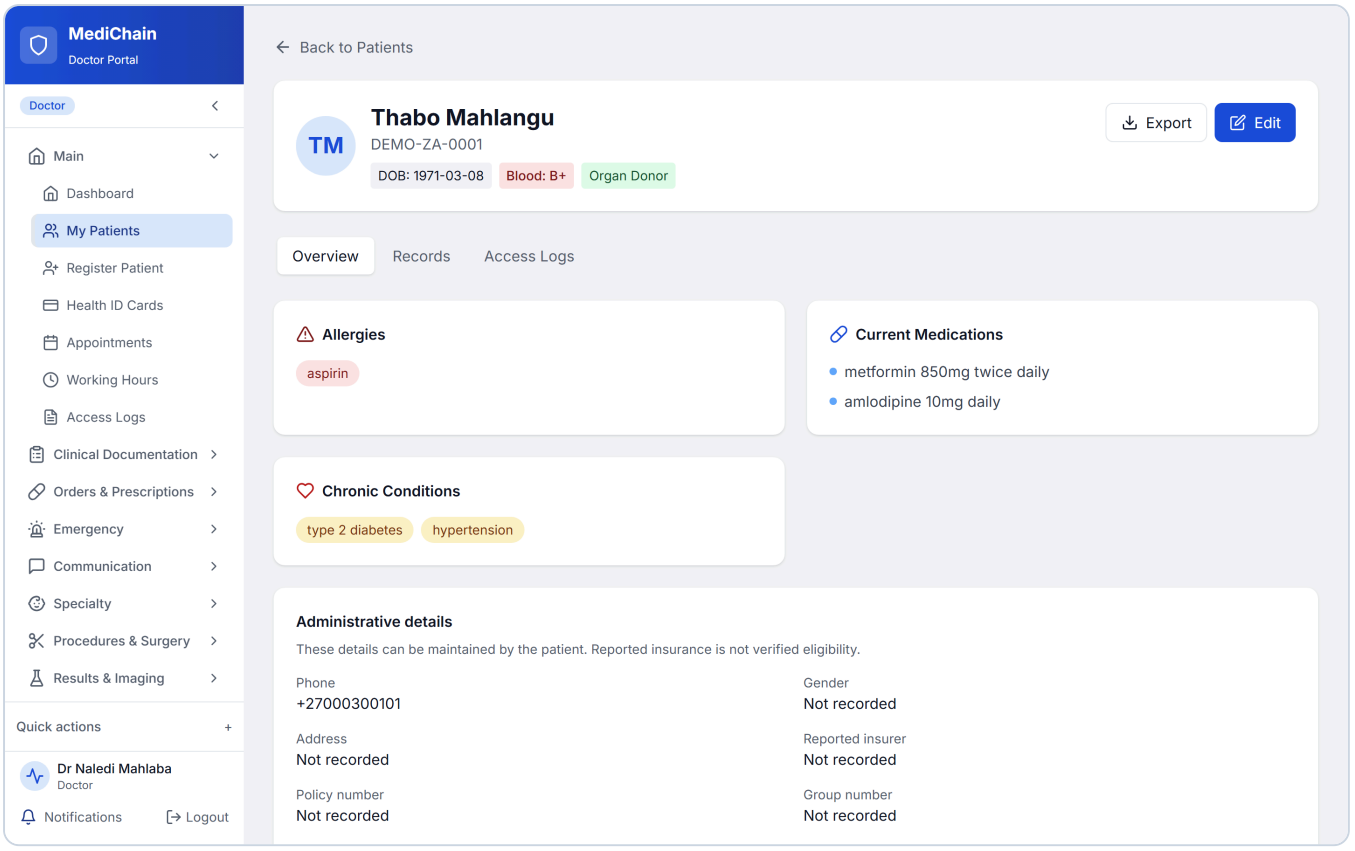


Consent and access grants

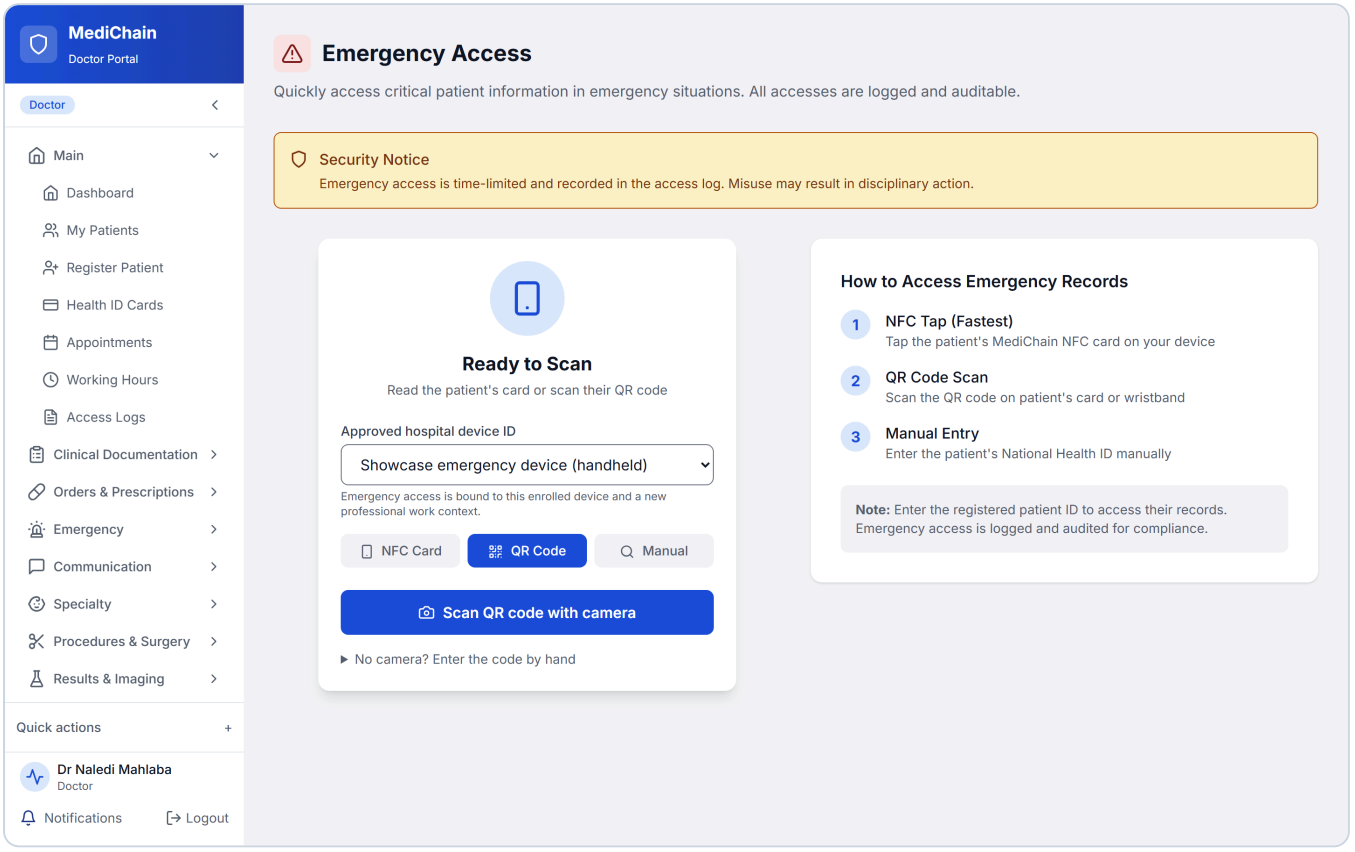


Staff portal: emergency access on a phone

Staff portal



Patient chart (Thabo Mahlangu)



Emergency access: QR scan, NFC card or card number